

A Conceptual Re-framing of Sensory Mismatch for Cybersickness

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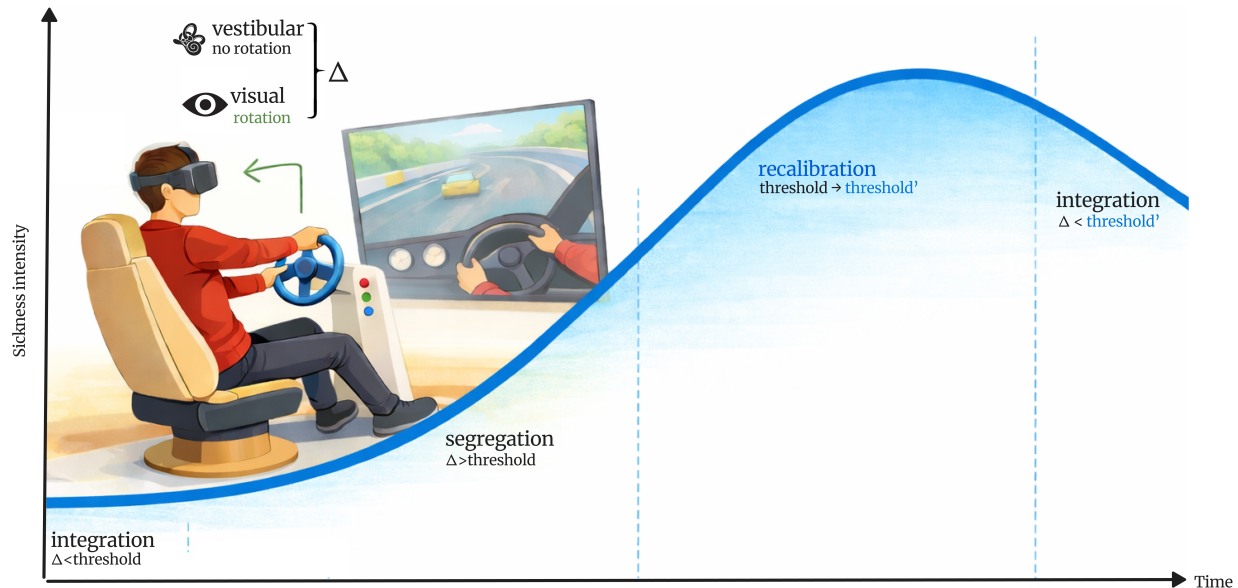


Figure 1: Sickness within multisensory perception. Visual and vestibular signals are initially integrated through reliability-based reweighting (integration, $\Delta < \text{threshold}$), where weights are dynamically adjusted according to momentary signal reliability. As discrepancies between visual and vestibular signals (Δ) increase beyond a threshold/disparity level (segregation, $\Delta > \text{threshold}$), they get segregated and do not provide a unified percept anymore. After a persistent disparity, recalibration gradually realigns the visual and vestibular signals into a unified percept again (recalibration, threshold $\rightarrow$ threshold'). Together, these processes shape the temporal dynamics of sickness, determining when symptoms emerge, how they progress during exposure, and how they subside.

Abstract

Despite two decades of hardware improvements, cybersickness remains a major barrier to the widespread and sustained adoption of virtual reality (VR). This paper proposes a conceptual re-framing of sensory mismatch, interpreting it as a quantifiable outcome of well-established cognitive mechanisms of multisensory perception,



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namely reliability-based cue reweighting, Bayesian causal inference, and cross-modal recalibration, which normally support perceptual stability and accurate self-motion estimation in natural environments. We argue that these mechanisms govern the onset, intensity, inter-individual variability, and post-exposure after-effects of sickness in VR. This re-framing has important implications for future VR research and design, offering a principled basis for developing perceptually coherent interfaces aimed at reducing cybersickness and for more clearly distinguishing visually induced motion sickness from other forms of sickness in VR.

CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models**; **Virtual reality**.

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1 Introduction

After significant technical advances in HMDs, *cybersickness* has become one of the strongest barriers to prolonged and widespread VR use. The cybersickness *symptoms* (visual fatigue, headache, pallor, sweating, dry mouth, stomach fullness, disorientation, vertigo, ataxia, nausea, and general tiredness) [25] do not only limit user experience but also constrain the design of VR scenes (e.g., limited locomotion and field of view) and the internal validity of VR-based studies in HCI.

Importantly, these symptoms did not emerge with VR itself. Motion sickness was described in the scientific literature as early as the 1930s, long before VR was invented. In the 1970s, Reason proposed the *sensory mismatch model* to explain the nature of motion sickness. Fifty years later, his model remains the most widespread in HCI, though it falls far short of addressing the exploratory challenges posed by the HCI community, such as prediction of sickness intensity, individual differences or sickness mitigation strategies. This leads to poorly justified experimental designs and weak, unreproducible results. In this work, **we argue that the sensory mismatch model alone cannot adequately account for the complexity of the phenomenon of cybersickness. Instead, we propose to treat sensory mismatch within a multisensory perception framework**, thoroughly studied in cognitive science during the last 20 years and possessing explanatory and predictive power [1, 8, 10, 12, 15, 17, 29–31].

This re-framing has both theoretical and practical implications for HCI. From a theoretical perspective, we will obtain a precise understanding of the conditions under which sickness emerges and does *not* emerge, as well as why and when its intensity increases or decreases. From a practical perspective, it enables us to design VR environments in a way that proactively minimize symptoms and, in the future, to develop genuine “VR recalibration training” for a smooth, seamless experience in VR.

In the following sections, we will discuss why the current description of sensory mismatch does not fully address the challenges

faced by the field, and why it nevertheless remains dominant over alternative accounts. We then examine the perceptual processes that happen in VR from the perspective of cognitive psychophysiology, briefly review the models used to describe these processes, and place sensory mismatch within this theoretical framework. Finally, we will briefly discuss the prospects of this approach and demonstrate how they can be operationalized in HCI in practice. We conclude with a concise roadmap and invite discussion during the poster session.

Multisensory integration is the process by which the brain merges signals from different senses into a single percept. For example, when seeing and holding a pen, signals from the visual and tactile modalities are combined into one coherent representation of the pen.

Reliability-based reweighting: If signals are not perfectly matched, the more reliable signal contributes more. For example, in a dark room, tactile information dominates over vision.

Causal inference: Whether signals originate from the same source is continuously evaluated. If the discrepancy becomes too large, e.g., when a TV image and speech are out of sync, the signals are no longer integrated, and the illusion that the speech is produced by the people breaks down.

Recalibration: When the discrepancy persists, sensory signals gradually adjust to each other, enabling integration once again.

2 Limits of the Sensory Mismatch Account

The sensory mismatch posits that *motion sickness* arises when there is a mismatch between the sensory input the brain expects (based on internal models of self-motion) and the actual sensory signals arriving from the visual, vestibular, and proprioceptive systems [20]. The classic example of motion sickness caused by sensory mismatch is reading in a moving car: the vestibular system detects acceleration and movement, while the visual system is focused on a stationary page and provides no corresponding motion cues. The ubiquity of the example, coupled with the explanatory simplicity of the model, made the sensory mismatch so popular that over time it acquired the status of a theory, although it is not [25, 26]. First, sensory mismatch has **weak predictive power**: it predicts that conflicting sensory inputs will cause nausea, but cannot predict its intensity or individual susceptibility. Second, it is **unfalsifiable**: any kind of motion sickness can be explained as a sensory mismatch. Third, the underlying concepts are **unmeasurable**, which makes it impossible to access the sickness intensity directly (instead, subjective scales of sickness, such as Fast Motion Sickness Scale (FMS) [14] or Simulator sickness questionnaire (SSQ) [13], are used along with sensors).

As a result, the model is interpreted inconsistently across studies, which can lead to mixed findings (see table 1 for details), flawed theoretical assumptions (e.g., that auditory cues can modulate the visuo-vestibular match [19]) and confusion in interpreting results (e.g., equating text motion on a static background with reading in a moving car [33]). Ultimately, this results in the term *cybersickness* becoming an umbrella for a broad range of symptoms that might have distinct origins (for example, visual fatigue or neck pain arise in the absence of visually-induced motion [25]). Further, we will use the term *visually induced motion sickness* (VIMS) to highlight the exact condition which sensory mismatch aims to explain.

Factor	Mixed Findings	Reference
Motion sickness susceptibility	High baseline motion-sickness susceptibility generally predicts higher cybersickness, but effect sizes range from strong to negligible;	[23, 25, 27]
Field-of-view (FOV)	Many studies report higher sickness with wide FOV, but others find minimal or no effect;	[5, 18, 23]
Locomotion type: continuous movement vs teleportation	Continuous locomotion (forward motion + rotations) typically provokes more motion-sickness symptoms than teleportation, but some studies (short or low-intensity exposures) show negligible differences between the two;	[2, 5, 23]
Vection strength	Strong vection is often associated with stronger motion sickness, but there are cases of high vection with limited sickness;	[5, 16, 24]
Exposure duration	Motion-sickness symptoms typically increase over time, but trajectories differ (monotonic increase, plateau, or adaptation).	[5, 23]

Table 1: Motion-related factors in VR with mixed empirical findings across studies

One potential reason why sensory mismatch remains popular is because all alternative explanations are even more vague and under-specified. For example, the evolutionary hypothesis argues that sickness evolves as a defense mechanism against neurotoxin ingestion, whereby abnormal patterns of neural activity are detected [28]. The ecological theory states the sickness arises due to the body’s inability to compensate for its posture given the external stimuli [21], although sickness might occur even without instability [7, 25]. The reweighting hypothesis by Gallagher and Ferrè proposes that visual and vestibular modalities do not solely provide matching or mismatching inputs, but are continuously evaluated and reweighted according to their reliability [8]. When visual input is less reliable, the vestibular modality dominates; when vestibular information is ambiguous, visual modality dominates. However, the description of how to measure visual and vestibular weights is vague, causing insufficient justification underlying the experimental designs and weak effects in results [4, 6].

Another potential reason why sensory mismatch remains popular is that it may still be grounded in a fundamentally correct principle. If framed within a strong theoretical framework, research on cybersickness could rapidly advance to a new level. An interdisciplinary perspective reveals that while the HCI community has primarily focused on mitigating cybersickness symptoms, cognitive science has developed well-established theoretical framework that is directly applicable to VIMS. In particular, research on *multisensory processing* has examined how signals from different sensory modalities are processed. While these insights were not originally developed in the context of VR, the sensory mismatch naturally aligns with them and provides a familiar conceptual bridge for CHI researchers studying cybersickness. In the next section, we quickly review them with a focus on how they align with the VR domain.

3 Sensory Mismatch within Multisensory Processing

Multisensory perception (so-called multisensory processing) refers to the fundamental property of the perceptual system to simultaneously receive and process information arising from multiple sensory modalities (vision, audition, touch, proprioception, vestibular, etc.). In everyday conditions, all modalities generate concurrent inputs across different sensory channels: an approaching object is simultaneously seen and heard; a grasped pen provides concurrent visual, haptic, and proprioceptive signals; self-motion produces

coordinated visual (optic flow) and vestibular cues. Meanwhile, the signals about one object do not remain segregated after processing: the brain compares the inputs from different modalities, evaluates the relative reliability of each sensory cue and combines them into a single, more accurate and robust perceptual *object representation*. This process of fusing heterogeneous sensory signals into a unified percept that surpasses the quality and precision of any individual modality is known as *multisensory integration*. It is precisely through multisensory integration that perceptual performance gains are achieved in accuracy, speed, and overall robustness in perceiving the surrounding world.

The brain continuously integrates visual, vestibular, and proprioceptive cues to maintain a stable percept for self-motion [1, 17]. In natural environments, information from visual and vestibular modalities is tightly coupled; however, in VR, physical motion is often absent, while optic flow is simulated. Current models of multisensory perception distinguish two independent processes that serve to resolve discrepancies between visual and vestibular modalities if they are not matched: fast dynamic reweighting of sensory cues according to their momentary reliability [10] and slow recalibration that remaps modalities toward each other [30] in case the discrepancies are too big to be considered as coming from the same source.

3.1 Visuo-vestibular Integration

When estimating head position in space, our perception integrates information primarily from visual and vestibular modalities in a statistically optimal manner following the principle of maximum likelihood estimation (MLE) [8]. Each modality provides an independent estimate of the *same* underlying state (e.g., head yaw angle), but with different levels of uncertainty. The *integrated estimate* is formed as a weighted average of the individual sensory estimates. The contribution of each modality to the combined estimate is determined by its reliability [10, 12]. More reliable sensory signals exert greater influence on the final estimate, whereas less reliable signals contribute less.

In visuo-vestibular integration experiments, varying visual reliability produces a smooth and predictable dynamic redistribution of visual and vestibular weights on a fast trial-to-trial basis [10]. If *only a visual stimulus* is present, the lower visual coherence leads to higher visual discrimination thresholds. If *visual and vestibular stimuli* are present as conflicting cues (1) in a high visual coherence condition the visual thresholds are minimal, and the final estimate

relies heavily on vision, while (2) at very low visual coherence it relies mainly on vestibular cue.

This in general goes in line with the reweighting hypothesis [11]. However, in visuo-vestibular integration experiments on heading discrimination, the conflict between visual and vestibular cues was kept *very small*, around 2–6° depending on stimuli reliability [10]. This setup ensures that participants perceive the visual and vestibular signals as coming from a *single common source*. At large conflicts ($\Delta \geq 20\text{--}30^\circ$) the percept *segregates* the visual and vestibular cues, according to the *Bayesian causal inference* (CI) model [1, 15, 22]. Unlike the MLE model, CI keeps uncertainty about the underlying *causal structure* given the sensory inputs [15] $C \in \{1, 2\}$, where $C = 1$ is a single common cause generating both cues (e.g., IVE self-motion producing congruent optic flow and vestibular signals) and $C = 2$, where the cues have independent causes (e.g., the observer is physically stationary while the visual scene rotates). When yaw difference is small, the probability of common cause $p(C = 1)$ remains high, and cues get integrated. When yaw difference exceeds the threshold/*disparity level*, the brain abruptly segregates the modalities ($C = 2$), and the VR user starts explicitly perceiving the visual scene as rotating while feeling physically stationary.

3.2 Visuo-vestibular Recalibration

When there is a long-term systematic misalignment between modalities, reweighting solely cannot fix it. Instead, another process of *multisensory recalibration* (adaptation, realignment) is launched to gradually adjust the baseline of each sensory modality to reduce persistent discrepancies between them and, therefore, bring the system back into balance [30]. The direction and magnitude depend on the discrepancy between the modality's own estimate and that of the other modality, scaled by its relative influence on the combined percept. From an evolutionary perspective, multisensory recalibration represents a crucial adaptive mechanism of neural plasticity aimed at maintaining both external accuracy (perceptual estimates should correspond to the true state of the world) and internal consistency (visual, vestibular, proprioceptive, and other cues should agree with one another) in a changing environment [17, 31]. Whereas fast dynamic reweighting is beneficial if the misalignment is small and of brief duration, in a long-term perspective integrating the two inputs, rather than ignoring one, confers a benefit in terms of variance: the integrated estimate has a smaller variance than either uni-modal estimate [3].

3.3 Multisensory Processing in VR

In the VR context, the rapid nature of reweighting cannot explain why the user does not immediately adapt to altered sensory conditions, nor does it account for the persistent after-effects observed after removing the headset. In turn, recalibration is a long-term shift in perception following a *systematic misalignment of signals* [31]. Although mechanisms of visuo-vestibular integration, causal inference, and recalibration have been extensively studied over the last 25 years [1, 9, 10, 12, 17, 29–31], their contribution to the emergence and mitigation of VIMS remains underexplored. At the same time, Reason's sensory mismatch (by definition) begins at the point where, due to a large discrepancy between visual and vestibular modalities, the two sensory inputs cease to be integrated.

Thus, it might be naturally embedded into multisensory perception framework.

4 Why Framing via Multisensory Processing Is The Way Forward

By embedding mismatch model into the multisensory perception framework, we no longer treat VIMS as a black box, but instead apply the multisensory processing toolkit to its study. In particular, this allows us to directly address the following questions: (1) the onset of sickness: what is the psychophysical threshold after which a VR user starts to feel sick; (2) individual susceptibility to sickness: how it is determined by higher or lower sensitivity to visuo-vestibular disparity; (3) adaptation to sickness: how is the gradual alleviation of symptoms after prolonged exposure related to recalibration of the visuo-vestibular system to a novel environment, and how does this recalibration persist or decay after the user removes the headset.

Below, we highlight three key implications that directly affect research methodologies, interpretation of results, and interaction design in virtual reality.

A direct measure to quantify sickness within a VR session

In traditional HCI research, VIMS is most often measured using questionnaires administered *before* and *after* a VR session. Such an approach does not allow researchers to identify the specific moment at which sickness onset occurs, nor to capture the dynamics of sickness over the course of the session. In the proposed approach, we hypothesize that VIMS reflects a transition from multisensory integration to multisensory segregation, governed by Bayesian causal inference. This transition can be operationalized by systematic increase of visuo-vestibular discrepancy and identifying the point at which participants report that visual and vestibular cues no longer originate from the same motion. This implies that the key dependent variable is defined as a psychophysical measure (the specific moment at which a threshold is crossed), rather than an indirect sickness rating after the VR session. This can be implemented by embedding brief discrimination session, e.g. an adaptive staircase procedure [32].

Disentangling different types of sickness in VR As cyber-sickness is used as an umbrella label for a wide range of subjective experiences from nausea and disorientation to visual fatigue and cognitive overload, it leads to a conceptual “flattening” of distinct phenomena and complicates the interpretation of experimental data. The proposed framework makes it possible to disentangle several mechanisms of discomfort: VIMS related to disruptions in visuo-vestibular integration; visual fatigue arising from the vergence-accommodation conflict (VAC); cognitive overload; and musculoskeletal discomfort. This separation enables more precise selection of dependent variables, more accurate interpretation of user reports, and more meaningful comparison of results across studies.

From Minimizing Motion to Supporting Stable Perceptual Inference in VR Design Currently, the strategy of minimizing sensory conflict still dominates: reducing motion, field of view, or visual complexity is assumed to lower the likelihood of VIMS. However, this strategy often results in overly constrained and impoverished forms of interaction. Viewing VIMS as a consequence of unstable

visuo-vestibular integration allows for a rethinking of VR interface design. Under this view, the critical factor is not the amount of motion, but instead the amount and rate of instances in which the visuo-vestibular discrepancy crosses the threshold/disparity level in opposite directions. The practical takeaway is that VR interaction design should focus on maintaining stable integrated percepts, rather than simply suppressing movement. Although it is still too early to propose specific design solutions, this perspective lays the groundwork for developing more effective mitigation strategies in the near future.

5 Research agenda

To begin with, we propose to conduct a series of experiments that combine the stages of visuo-vestibular processing while measuring VIMS in parallel. A promising starting point would be to systematically manipulate the coherence of visual motion cues (e.g., optic flow patterns) to vary the reliability of visual cues. This would make it possible to examine how changes in sensory reliability shift the balance between visuo-vestibular integration and segregation, and whether crossing this balance point predicts the emergence of VIMS symptoms. In the context of the poster session, we invite the HCI community to discuss:

- (1) Which VR environments are most appropriate for the first experiment (e.g., more naturalistic driving simulations or more controlled random dot-kinematograms);
- (2) The prospects and limitations of using physiological sensors and models (e.g., electrodermal activity/skin conductance (EDA/SC), eye tracking, including pupil position and blink rate, head motion, etc.);
- (3) How the theoretical concepts and measures developed in cognitive science should be communicated in a way that makes them most useful and actionable for the HCI community.

We hope that such a dialogue will help to establish a shared experimental language between cognitive science and HCI and will constitute a step toward more theoretically grounded, reproducible, and predictive research on cybersickness in VR.

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